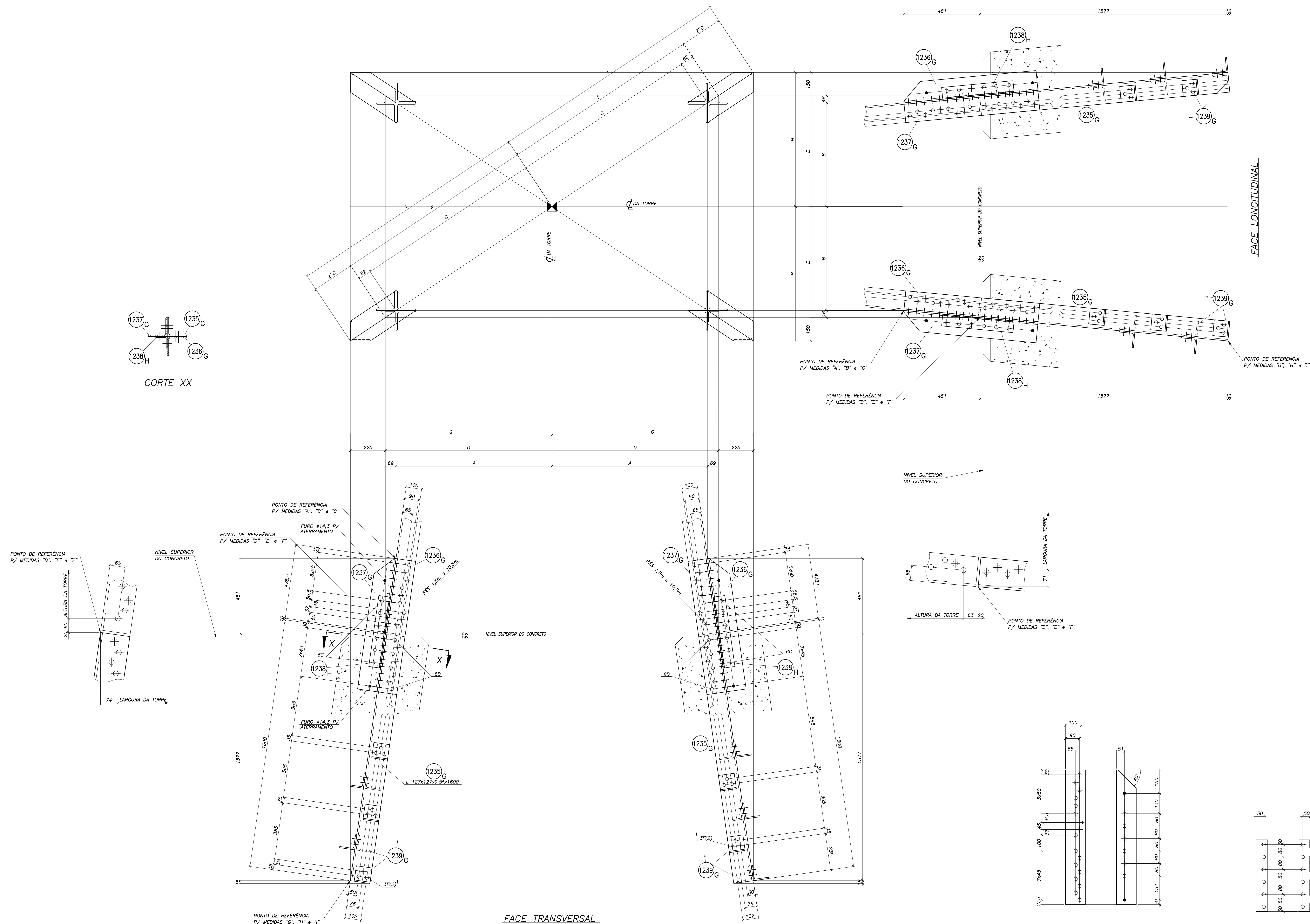
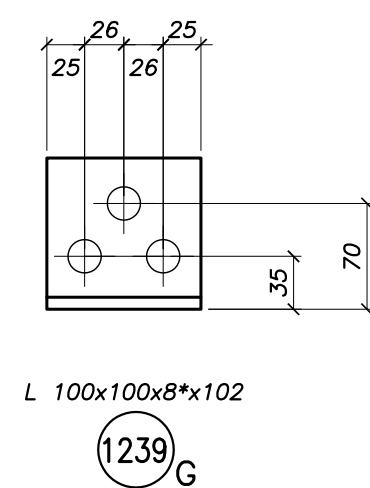
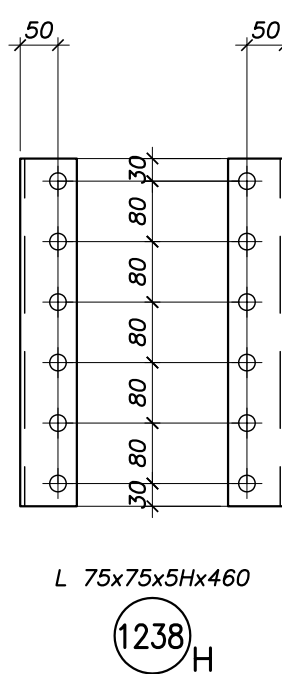
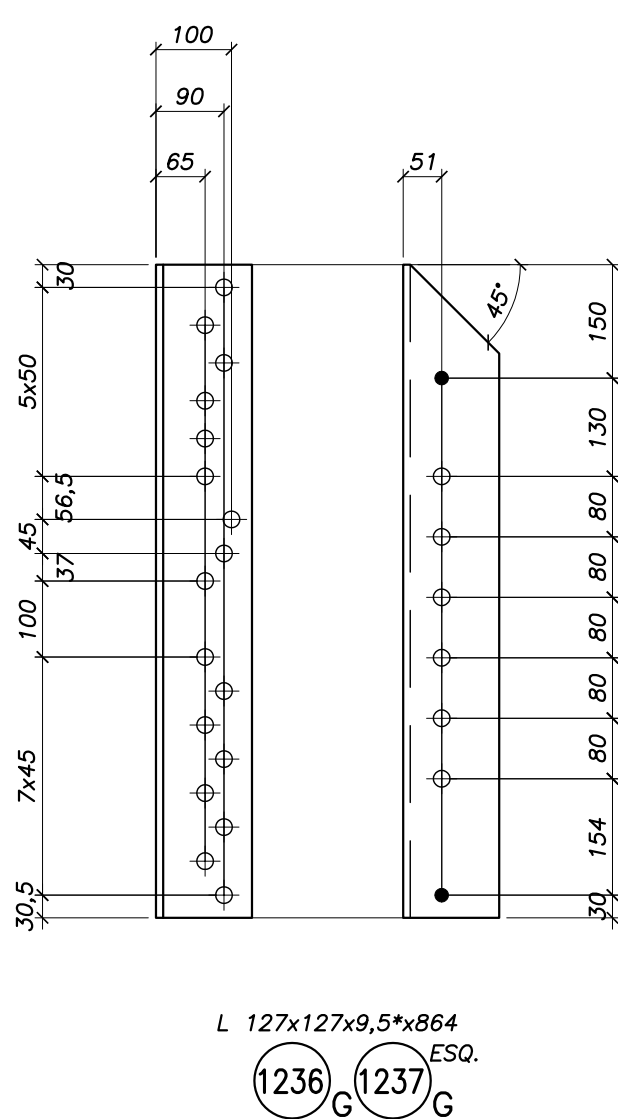
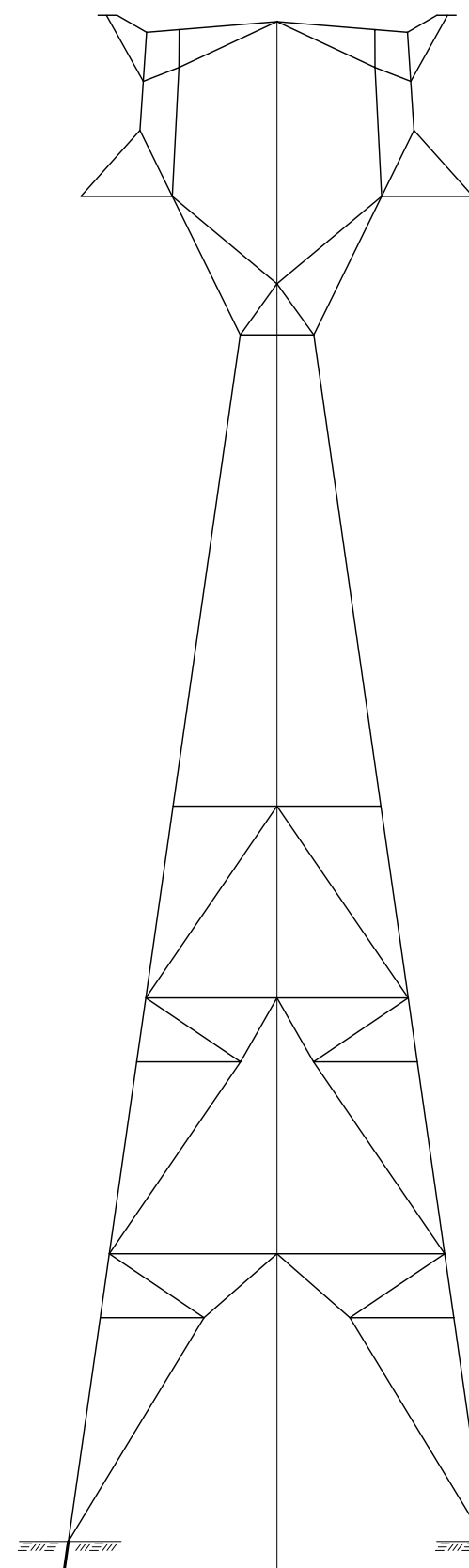


| Rev. | Data     | Descrição          | Por | Aprov. |
|------|----------|--------------------|-----|--------|
| 04   | 13/10/21 | Emissão Inicial    | FGS | PRRLS  |
| 0    | 27/01/22 | Aprovado           | NLA | PRRLS  |
| 1A   | 18/07/22 | Conforme Fabricado | WHC | PRRLS  |
| 1    | 19/07/22 | Aprovado           | PCM | PRRLS  |
|      |          |                    |     |        |



| TORRES                 | PES   | A    | B    | C     | D    | E    | F     | G     | H    | I     |
|------------------------|-------|------|------|-------|------|------|-------|-------|------|-------|
| TORRE C/<br>EXT. 0.0m  | 1.5m  | 5094 | 3418 | 6134  | 5162 | 3463 | 6216  | 5387  | 3613 | 6486  |
|                        | 3.0m  | 5307 | 3560 | 6390  | 5376 | 3606 | 6483  | 5601  | 3756 | 6744  |
|                        | 4.5m  | 5521 | 3703 | 6648  | 5590 | 3748 | 6730  | 5814  | 3898 | 7000  |
|                        | 6.0m  | 5735 | 3845 | 6905  | 5803 | 3891 | 6987  | 6028  | 4041 | 7257  |
|                        | 7.5m  | 5949 | 3988 | 7162  | 6017 | 4033 | 7244  | 6242  | 4183 | 7514  |
|                        | 9.0m  | 6162 | 4130 | 7418  | 6231 | 4176 | 7501  | 6456  | 4326 | 7771  |
| TORRE C/<br>EXT. 6.0m  | 10.5m | 6376 | 4273 | 7675  | 6445 | 4318 | 7758  | 6669  | 4468 | 8027  |
|                        | 1.5m  | 5949 | 3988 | 7162  | 6017 | 4033 | 7244  | 6242  | 4183 | 7514  |
|                        | 3.0m  | 6162 | 4130 | 7418  | 6231 | 4176 | 7501  | 6456  | 4326 | 7771  |
|                        | 4.5m  | 6376 | 4273 | 7675  | 6445 | 4318 | 7758  | 6669  | 4468 | 8027  |
|                        | 6.0m  | 6590 | 4415 | 7932  | 6658 | 4461 | 8014  | 6883  | 4611 | 8285  |
|                        | 7.5m  | 6804 | 4558 | 8190  | 6872 | 4603 | 8271  | 7097  | 4753 | 8542  |
| TORRE C/<br>EXT. 12.0m | 9.0m  | 7017 | 4700 | 8446  | 7086 | 4746 | 8529  | 7311  | 4896 | 8799  |
|                        | 10.5m | 7231 | 4843 | 8703  | 7300 | 4888 | 8785  | 7524  | 5038 | 9055  |
|                        | 1.5m  | 6804 | 4558 | 8190  | 6872 | 4603 | 8271  | 7097  | 4753 | 8542  |
|                        | 3.0m  | 7017 | 4700 | 8446  | 7086 | 4746 | 8529  | 7311  | 4896 | 8799  |
|                        | 4.5m  | 7231 | 4843 | 8703  | 7300 | 4888 | 8785  | 7524  | 5038 | 9055  |
|                        | 6.0m  | 7445 | 4985 | 8960  | 7513 | 5031 | 9042  | 7738  | 5181 | 9312  |
| TORRE C/<br>EXT. 18.0m | 7.5m  | 7659 | 5128 | 9217  | 7727 | 5173 | 9299  | 7952  | 5323 | 9569  |
|                        | 9.0m  | 7872 | 5270 | 9473  | 7941 | 5316 | 9556  | 8166  | 5466 | 9827  |
|                        | 10.5m | 8086 | 5413 | 9731  | 8155 | 5458 | 9813  | 8379  | 5608 | 10083 |
|                        | 1.5m  | 7659 | 5128 | 9217  | 7727 | 5173 | 9299  | 7952  | 5323 | 9569  |
|                        | 3.0m  | 7872 | 5270 | 9473  | 7941 | 5316 | 9556  | 8166  | 5466 | 9827  |
|                        | 4.5m  | 8086 | 5413 | 9731  | 8155 | 5458 | 9813  | 8379  | 5608 | 10083 |
| TORRE C/<br>EXT. 24.0m | 6.0m  | 8300 | 5555 | 9987  | 8368 | 5601 | 10069 | 8593  | 5751 | 10340 |
|                        | 7.5m  | 8514 | 5698 | 10245 | 8582 | 5743 | 10326 | 8807  | 5893 | 10597 |
|                        | 9.0m  | 8727 | 5840 | 10501 | 8796 | 5886 | 10584 | 9021  | 6036 | 10854 |
|                        | 10.5m | 8941 | 5983 | 10758 | 9010 | 6028 | 10841 | 9234  | 6178 | 11110 |
|                        | 1.5m  | 8514 | 5698 | 10245 | 8582 | 5743 | 10326 | 8807  | 5893 | 10597 |
|                        | 3.0m  | 8727 | 5840 | 10501 | 8796 | 5886 | 10584 | 9021  | 6036 | 10854 |
| TORRE C/<br>EXT. 30.0m | 4.5m  | 8941 | 5983 | 10758 | 9010 | 6028 | 10841 | 9234  | 6178 | 11110 |
|                        | 6.0m  | 9155 | 6125 | 11015 | 9233 | 6171 | 11097 | 9448  | 6321 | 11367 |
|                        | 7.5m  | 9369 | 6268 | 11272 | 9437 | 6313 | 11354 | 9662  | 6463 | 11624 |
|                        | 9.0m  | 9582 | 6410 | 11528 | 9651 | 6456 | 11611 | 9876  | 6606 | 11882 |
|                        | 10.5m | 9796 | 6553 | 11786 | 9865 | 6598 | 11868 | 10089 | 6748 | 12139 |



### CONVENÇÃO DE RECORTE

Diagram illustrating the convention for cutting screws. It shows five types of screw heads: hexagonal (A), hex flange (B), hex washer head (C), hex cap (D), and hex button (E). Each type is shown in a side view and a top view. The side view shows the head profile and the shank. The top view shows the head shape and the shank diameter. The labels A, B, C, D, and E are placed next to the corresponding screw head.

### PARAFUSOS

| Símbolo                | Parafuso              | Ø Furo | K  | R  | COMPRIMENTOS | TORQUE (kgf.m) |
|------------------------|-----------------------|--------|----|----|--------------|----------------|
|                        | M12                   | 13,5   | 20 | 22 | A=30<br>H=65 | Ø Min. Max.    |
|                        | M16                   | 17,5   | 25 | 26 | B=35<br>I=70 | M12 3,0 4,5    |
|                        | M20                   | 21,5   | 30 | 32 | C=40<br>J=75 | M16 7,5 11,5   |
|                        | M24                   | 25,5   | 35 | 34 | D=45<br>K=85 | M20 13,0 22,0  |
|                        | Parafuso Degrau Ø15,9 | 17,5   | -  | 70 | F=50<br>L=85 | M24 20,0 39,0  |
| X Quando não indicado: |                       |        |    |    | G=60         |                |
|                        |                       |        |    |    | F=60         |                |

### MONTAGEM

Diagram illustrating the assembly of a screw into a hole. It shows a screw being inserted into a hole in a plate. The plate has a thickness labeled 'ESPESSURAS'. A washer labeled 'PAINA' is placed between the screw head and the plate. A nut labeled 'PORCA' is placed on the other side of the plate. The diagram shows the screw being tightened into the nut.

| FACE TRANSVERSAL        |                          | FACE LONGITUDINAL       |                         |
|-------------------------|--------------------------|-------------------------|-------------------------|
| VERTICAL                | FACE                     | VERTICAL                | FACE                    |
| tg $\alpha$ = 0,142500  | tg $\varphi$ = 0,141861  | tg $\gamma$ = 0,095000  | tg $\omega$ = 0,094050  |
| 2tg $\alpha$ = 0,285000 | sec $\varphi$ = 1,010008 | 2tg $\gamma$ = 0,190000 | sec $\omega$ = 1,004408 |
| sec $\gamma$ = 1,004505 |                          | sec $\alpha$ = 1,010106 |                         |
| sec $\beta$ = 1,014558  |                          |                         |                         |

1) REFERÊNCIA:  
1a) ESPECIFICAÇÕES E NOTAS GERAIS ..... 482.03-D1300.

|                      |  |    |  |    |  |    |  |    |  |    |  |    |  |    |  |    |  |          |  |        |  |     |  |                                 |  |                            |  |     |  |     |  |          |  |  |  |        |  |             |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|----------------------|--|----|--|----|--|----|--|----|--|----|--|----|--|----|--|----|--|----------|--|--------|--|-----|--|---------------------------------|--|----------------------------|--|-----|--|-----|--|----------|--|--|--|--------|--|-------------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
| POS. 1235 - 1239     |  |    |  |    |  |    |  |    |  |    |  |    |  |    |  |    |  |          |  |        |  |     |  | PROJETO EXECUTIVO ELTROMECANICO |  |                            |  |     |  |     |  |          |  |  |  |        |  |             |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| M16                  |  | -- |  | 12 |  | 16 |  | -- |  | 15 |  | -- |  | -- |  | 58 |  | --       |  | --     |  | --  |  | 43                              |  | Escala 1:10, 1:5           |  |     |  |     |  |          |  |  |  |        |  | Folha 01/01 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| M12                  |  | -- |  | -- |  | -- |  | -- |  | -- |  | -- |  | -- |  | -- |  | --       |  | --     |  | --  |  | --                              |  | 1 Contratada 482.03-D13030 |  |     |  |     |  |          |  |  |  |        |  | Rev. 1      |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| 8                    |  | 30 |  | 35 |  | 40 |  | 45 |  | 50 |  | 55 |  | 60 |  | 65 |  | 70       |  | DEGRUS |  | 3.0 |  | 5.2                             |  | 3.2                        |  | 4.8 |  | 6.4 |  | PAC-NUTS |  |  |  |        |  |             |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| PARAFUSOS            |  |    |  |    |  |    |  |    |  |    |  |    |  |    |  |    |  | ARRUELAS |  |        |  |     |  |                                 |  |                            |  |     |  |     |  |          |  |  |  | CALÇOS |  |             |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| QUANTIDADES P/ 1 SUB |  |    |  |    |  |    |  |    |  |    |  |    |  |    |  |    |  |          |  |        |  |     |  |                                 |  |                            |  |     |  |     |  |          |  |  |  |        |  |             |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| N° EKT 7             |  |    |  |    |  |    |  |    |  |    |  |    |  |    |  |    |  |          |  |        |  |     |  | LT-L-GERAL-ES-A0-3134           |  |                            |  |     |  |     |  |          |  |  |  |        |  |             |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

|                                                                                                                                                                                                                                           |                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
|                                                                                                                                                      |                                                 |
| Projeto                                                                                                                                                                                                                                   | PWV 13/10/21                                    |
| Verificação                                                                                                                                                                                                                               | FSG 13/10/21                                    |
| Permissão                                                                                                                                                                                                                                 | ROM 13/10/21                                    |
| Responsável Técnico                                                                                                                                                                                                                       | PRRLS 13/10/21<br>Carg. c/ab. DIRETOR MG19042/D |
| <b>EXCIT 7</b><br>Serviço de Transmissão de Energia Elétrica - CTE 4                                                                                                                                                                      |                                                 |
| <br><b>NEOENERGIA</b>                                                                                                                                |                                                 |
| Nome da Obra: <b>LEILÃO ANEL 01/2020 - LOTE 2</b><br><b>LINHAS DE TRANSMISSÃO 500 KV CIRCUITO SIMPLAS</b><br><b>MORRO DO CHAPELÉ U / POÇÕES C1</b><br><b>POÇÕES II / MEDEIROS NETO II C1</b><br><b>MEDEIROS NETO II / JOÃO NEIVA 2 C1</b> |                                                 |
| Título do Documento:                                                                                                                                                                                                                      |                                                 |
| <h1>TORRE TIPO N51SL</h1> <h2>STUB</h2>                                                                                                                                                                                                   |                                                 |
| Classificação do Projeto:                                                                                                                                                                                                                 |                                                 |
| <b>PROJETO EXECUTIVO ELETROMECÂNICO</b>                                                                                                                                                                                                   |                                                 |
| Escala: 1:10, 1:5                                                                                                                                                                                                                         |                                                 |
| Folha: 01/01                                                                                                                                                                                                                              |                                                 |
| Nº Contratada: 482.03-D13030 Rev. 1                                                                                                                                                                                                       |                                                 |
| Nº EXCIT 7 LT-L-GERAL-ES-AO-3134                                                                                                                                                                                                          |                                                 |